

2% if nothing happens and then 2% plus 20% when the performance fee kicks in. This payoff is highly convex—essentially the mirror image of most HF strategies, which are concave. That is, HF compensation contracts are *long call options* and long volatility, while the strategies employed by HF managers are short put options and short volatility.

The performance fees, as a sequence of call options, encourage the HF manager to take risks.⁴¹ If they pay off, you can hit the big leagues and sit beside Dalio and Simons. If your bet doesn't work out, you don't lose anything and still get the 2% management fee. Your investors, of course, lose. In the worst case, you perform so badly you have to close up shop. After you fail, you open up a new HF (this is a *restart option* and is a form of *implicit contract*, see chapter 15). The former partners of the failed LTCM did just that. A year after LTCM failed in 1998, Myron Scholes set up Platinum Grove Asset Management, which suffered like many other HFs during 2008 and gated its investors.⁴² Scholes retired from Platinum Grove in 2011.⁴³ John Meriwether, another partner at LTCM, started JWM Partners with other ex-LTCM colleagues. It closed in 2009 after losses during the financial crisis.⁴⁴ Like the phoenix, new HFs also arose from the implosion of Goldman's quant funds after the quant meltdown and financial crisis. Mark Carhart, who co-headed Global Alpha, set up Kepos Capital, and Raymond Iwanowski, the other co-manager, opened Secor Asset Management.

The value of the option performance fee is roughly 3% to 4%, as first shown by Goetzmann, Ingersoll, and Ross (2003). In his PhD dissertation, my former student Sergiy Gorovyy (2012) estimated equivalent flat management fees (that gave the manager the same total compensation) for various performance structures. He found a straight 2/20 contract equivalent to a 6.4% management fee. Adding a high-water mark brought this down to 5.8%. A 2/20 contract with a high-water mark and a 4% hurdle rate was equivalent to a 5.3% management fee. The average active mutual fund fee is below 1%. These are high fees.

Asset owners shouldn't mind paying high fees but only when there is outperformance. Sadly, this is not the case in current HF contracts. Lan, Wang, and Yang (2012) build a model of HF leverage and valuation. They find that for the standard 2/20 contract, the HF manager needs to create a whopping 20% increase on

⁴¹ See Goetzmann, Ingersoll, and Ross (2003), Hodder and Jackwerth (2007), and Lan, Wang, and Yang (2012). Panageas and Westerfield (2003) argue that the case when the HF has an infinite horizon, but the fund has a probability of being terminated exogenously each period, reduces these risk-taking incentives.

⁴² Kishan, S., "Scholes's Platinum Grove Fund Halts Withdrawals After Losses," Bloomberg, Nov. 6, 2008.

⁴³ Comstock, C., 2011, "Founder Myron Scholes Retired From His Hedge Fund," Business Insider, Feb. 10, 2011.

⁴⁴ Burton, K., and S. Kishan, "Meriwether Said to Shut JWM Hedge Fund After Losses," Bloomberg, July 8, 2009.